

CONTACT

EMAIL
hsandmann@ieee.org
WEB
hsandmann.github.io
GITHUB
hsandmann
LINKEDIN
in/hsandmann
PROJECT
graphole.com
LOCATION
São Paulo, Brazil
RELOCATION
Open to relocation (on-site)

SKILLS

Languages

Java · Python · Go · Kotlin · C · C++ · JavaScript · SQL

Machine Learning & AI

Deep Learning · Neural Networks · Computer Vision · NLP · PCA · SVM · Clustering

Backend & Architecture

Spring Boot · Spring Cloud · Microservices · Design Patterns · Software Architecture

Data & Infrastructure

PostgreSQL · Redis · Kafka · RabbitMQ · Docker · Kubernetes · Spark · MongoDB · Neo4j

Domains

Banking & Fintech · Cryptography (PKI/HSM) · High-Performance Computing · POS/ERP

EDUCATION

D.Sc. · Electrical Engineering
Polytechnic School, USP · 2007–2012

M.Sc. · Electrical Engineering
Polytechnic School, USP · 2003–2006

B.Sc. · Computer Science
FEI · 1999–2002

LANGUAGES

Portuguese	Native
English	Advanced
German	B1

Humberto Sandmann

Computer Scientist & Professor

PROFILE

Senior software architect and artificial intelligence engineer with 20+ years building high-stakes systems across banking, fintech, and retail — core banking, cryptography (PKI/HSM), high-performance computing, computer vision, and cloud-native microservices. D.Sc. in neural computing; also a university professor teaching software design, AI, and distributed systems.

EXPERIENCE

Founder · H-Sandmann 2006–2025

Independent consultancy delivering short-term, specialized engagements in machine learning, optimization and security.

Machine Learning · Software Architecture

Co-Founder · SelsanTech 2012–2023

Co-founded a venture bringing computational-intelligence research to market; built ZigTrack, real-time indoor tracking of equipment with low-cost fixed sensors.

Machine Learning · Software Architecture

Senior Software Architect & Machine Learning Engineer · RockSpoon 2018–2019

Team leader and product owner; architected POS/ERP/payroll platforms and computer-vision/ML features for a restaurant-tech company.

POS · ERP · JavaEE · PostgreSQL · OpenCV · Computer Vision · Go · Android · Kotlin

Senior Consultant, Computational Optimization · Banco Votorantim 2015–2016

Return-on-equity computation for 2M+ records within a 12-hour window on high-performance computing infrastructure for a major bank.

RoE · High-Performance Computing · J2EE · Spring Boot · Jenkins · Test Automation

Senior Consultant, Computational Optimization · TecBan 2011

Senior Software Engineer / Architect · ABN Amro Bank 2006–2007

Senior System Analyst · Politec 2005–2006

Senior Software Engineer / Architect · EverSystems 2004–2005

Senior System Analyst · Central Online 2002–2003

Senior System Analyst · Datamace 1999–2002

Computer Programmer · Heraeus Electro-Nite 1998–1999

ACADEMIC POSITIONS

Associate Professor · Insper 2018–present

Associate Professor · ESPM 2014–2025

Associate Professor (freelance) · Link School of Business 2024

Professor · IBTA — Instituto Brasileiro de Tecnologia Avançada 2016–2017

Associate Professor · FIAP 2012–2015

Visiting Professor · Toledo Prudente Centro Universitário 2015

Associate Professor · Universidade Cruzeiro do Sul 2012–2013

Temporary Professor · Universidade Nove de Julho 2011

Teaching Assistant · Polytechnic School, University of São Paulo 2008–2011

Full academic CV, publications and teaching record at hsandmann.github.io